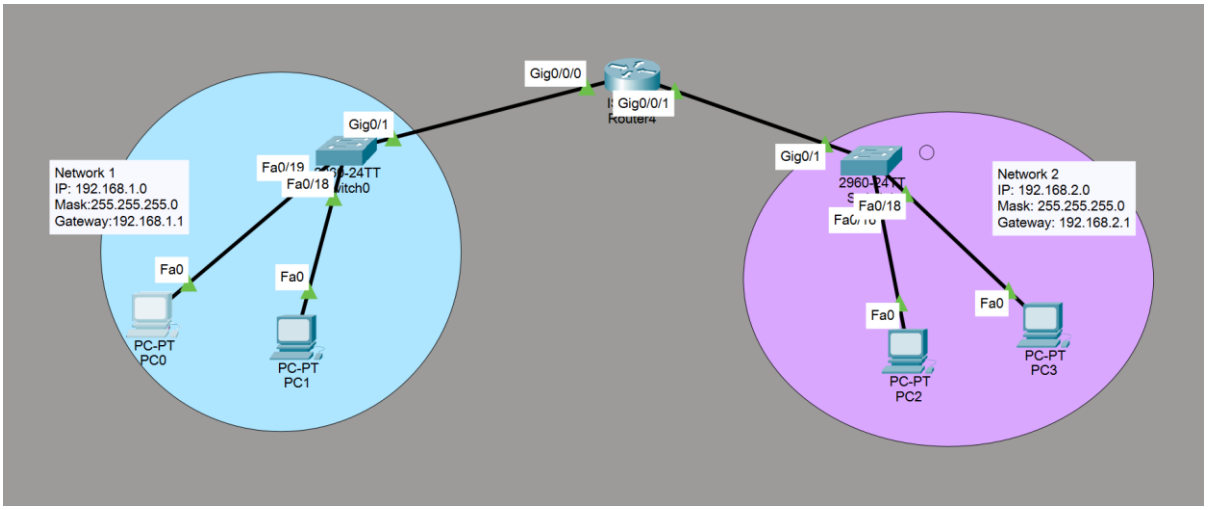
Part 1: Network Topology and Configuration

Topology:



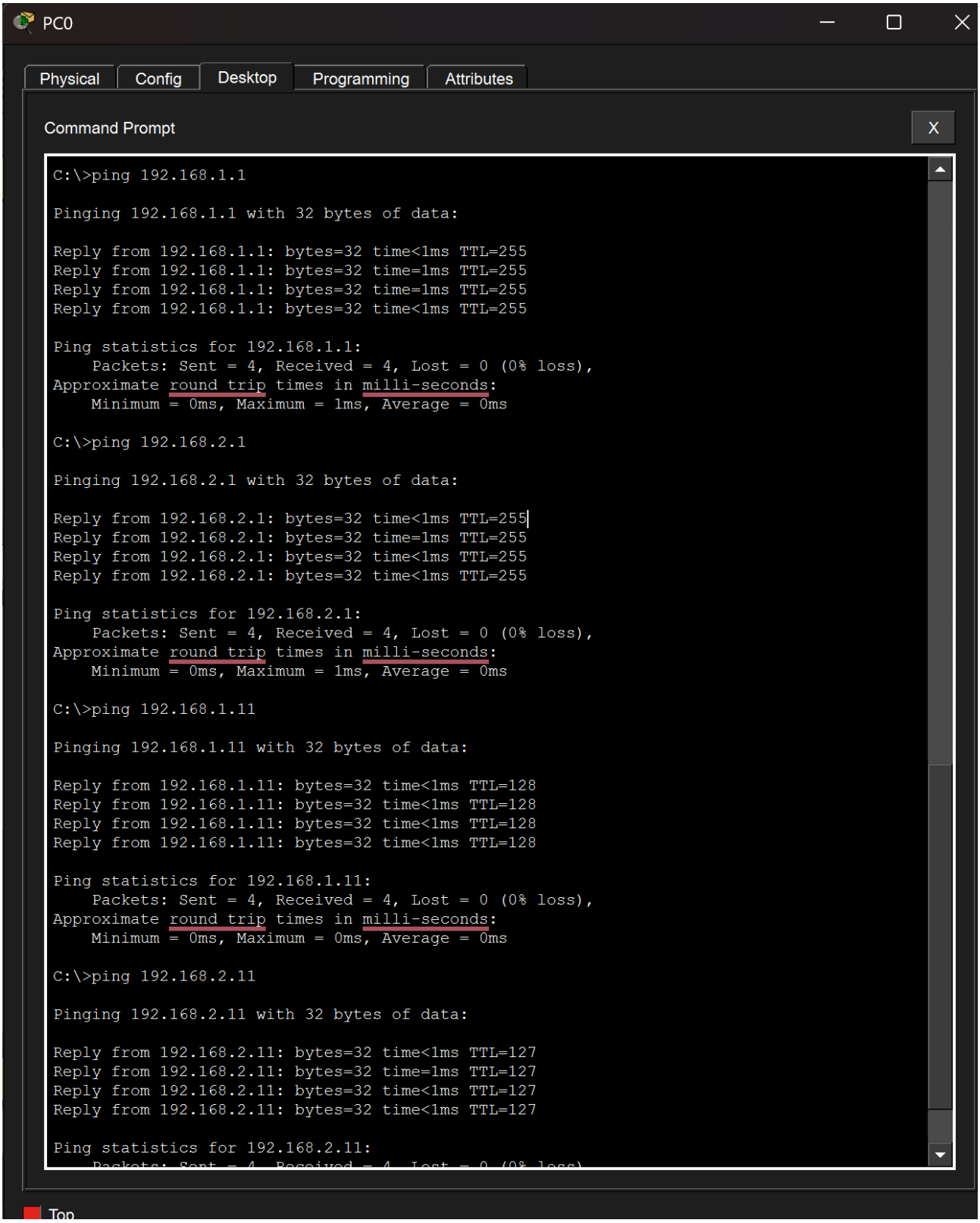
IP Addressing Table:

Network	IP Addressing	Mask	Gateway
N1	192.168.1.0	255.255.255.0	192.168.1.1
N2	192.168.2.0	255.255.255.0	192.168.2.1

Successful pinging:

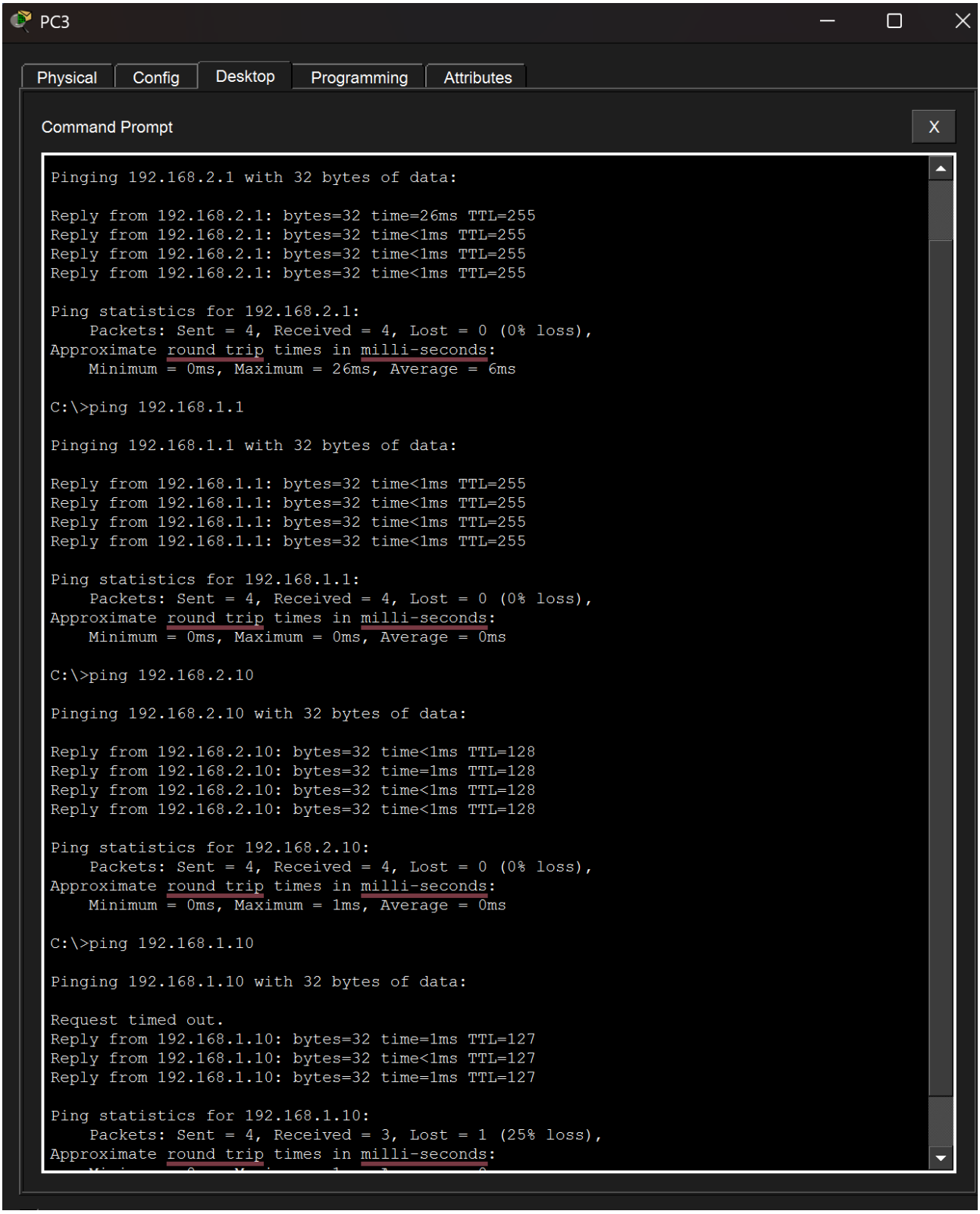
N1:

First ping to the router in the same network. Second ping to the router from a different network. Third ping to the PC in the same network. Fourth ping to the PC from a different network



N2:

First ping to the router in the same network. Second ping to the router from a different network. Third ping to the PC in the same network. Fourth ping to the PC from a different network



Explanation:

In this task, I created a network topology with two different networks (two separate subnets). The network includes one router, two switches, and four PCs, with two PCs in each network. The switches are used to connect the PCs in the same network, and the router is used to connect two networks together so that devices from different networks can communicate.

The first network uses the address 192.168.1.0 with a subnet mask of 255.255.255.0. The default gateway for this network is 192.168.1.1. PC0 and PC1 are connected to the first switch and were given IP addresses from this network, such as 192.168.1.10 and 192.168.1.11. Both PCs use 192.168.1.1 as their default gateway.

The second network uses the address 192.168.2.0 with a subnet mask of 255.255.255.0. The default gateway for this network is 192.168.2.1. PC2 and PC3 are connected to the second switch and were given IP addresses such as 192.168.2.10 and 192.168.2.11. Their default gateway is set to 192.168.2.1.

The router connects two networks. I configured two router interfaces. The first interface was given the IP address 192.168.1.1 to connect to Network 1, and the second interface was given the IP address 192.168.2.1 to connect to Network 2. The command "no shutdown" was used to activate interfaces.

I tested the connection using the ping command. All successful pinging commands were shown above. This confirmed that the router is correctly routing traffic and devices in two different networks can communicate with each other. As long as it was an easy task, no challenges occurred.

Part 1: OSI and TCP/IP Model

Task 1: Model Diagrams

OSI Model

Application Layer

What it does: This is the layer you see. It lets programs like web browsers and email apps use the network. It enables direct interaction between end-user applications and the network.

Example Protocol: Hypertext Transfer Protocol (HTTP) – used when you open a website.

Presentation Layer

What it does: It makes sure the data looks right. It can change the format, lock the data (encryption), or make it smaller (compression).

Example Protocol: Transport Layer Security (TLS) – keeps data safe and private.

Session Layer

What it does: It starts, manages, and ends the connection between two computers. It keeps the conversation organized.

Example Protocol: Remote Procedure Call (RPC) – helps one computer talk to another.

Transport Layer

What it does: It makes sure the data arrives correctly. If something is missing, it asks for it again.

Example Protocol: Transmission Control Protocol (TCP) – sends data safely and in order.

Network Layer

What it does: It decides the best path for the data to travel to reach the right computer.

Example Protocol: Internet Protocol (IP) – uses IP addresses to send data to the correct place.

Data Link Layer

What it Does: It sends data between devices on the same network

Example Protocol: Ethernet (IEEE 802.3) – used in wired local networks.

Physical Layer

What it does: It sends raw bits through cables, fiber, or wireless signals

Example Protocol: Universal Serial Bus (USB) – defines how devices physically connect.

TCP/IP Model

Application Layer

What it does: This layer allows programs like web browsers, email apps, and file transfer tools to communicate over the internet. It is where users interact with network services.

Example Protocol: Hypertext Transfer Protocol (HTTP) – used when you open a website.

Transport Layer

What it does: This layer makes sure data is delivered between devices. It can make sure the data arrives correctly and in the right order..

Example Protocol: Transmission Control Protocol (TCP) – sends data reliably between computers.

Internet Layer

What it does: This layer sends data packets between different networks and decides the path the data should take.

Example Protocol: Internet Protocol (IP) – uses IP addresses to deliver data to the correct device.

Network Access Layer

What it does: This layer handles how data is physically sent through the network, like through cables or wireless signals.

Example Protocol: Ethernet (IEEE 802.3) – used for communication in local networks.

Task 2: Encapsulation and Decapsulation

